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Digital Forensics

14x065

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0.1 Principles of digital forensics

Acquisition: power-off or write-block the machine, then bit-copy the evidence so time-stamps and volatile data are frozen.

Identification: surface every byte—recover slack space, steganography, encrypted volumes—so nothing that may later refute a claim is overlooked.

Evaluation: build a chronology (MAC times, registry keys, network flows) that links user action to artefact.

Presentation: translate the technical chain of custody into diagrams and sworn statements a jury or C-level executive can grasp.

Typical clients: criminal prosecutors, civil litigants, insurers, corporations hunting insider theft.

0.2 Digital multimedia forensics: definition, classification.

Ubiquitous cheap editing broke photography's age-old "seeing is believing".

We now treat every image, video, audio clip as *potentially* synthetic.

Societal cost: opinion polarisation, diminished trust in photojournalism, courtroom scepticism.

Scientific response: treat each file as a noisy measurement of the *entire* history from sensor through every post-process, then invert that history.

0.2.1 Idea

Both in-camera physics and later edits leave *intrinsic* artefacts—think of them as digital fingerprints.

Digital fingerprints

- **In-camera:** optical low-pass filter, CFA layout, sensor read-noise $\propto I_0(x, y)K(x, y)$ and firmware colour matrix leave a predictable signature.
- **Out-camera:** every re-save, resize, contrast stretch or Instagram filter nudges statistics (DCT histogram, bi-coherence, error-in-variables) away from the factory baseline.
- **Scene-geometric:** lighting direction L , shadow penumbra, eye-specular s obey projective constraints; mismatches betray compositing.

0.3 Active and passive approaches for multimedia authentication.

0.3.1 Active approach

- **Cryptographic signature:** sign a perceptual hash at capture; verify later with public key or blockchain anchoring. Needs PKI end-to-end.
- **Fragile/semi-fragile watermark:** embed energy in LSB or middle-band DCT; any change flips bits or lifts correlation.
Drawbacks: crypto dies at decryption, re-compression breaks bit-wise integrity, and watermarks must be inserted *before* forgery—rare in real life.

0.3.2 Passive and blind approach: multimedia forensics

No ex-ante mark required; examiner deduces *a posteriori* answers to:

- Who/what created this? (source)
- Was it cooked? (integrity)

Advantage: works on legacy data, no standards war, no key management.

0.4 Image generation pipeline

1. **Light projection:** irradiance $E(x, y)$ hits CFA.
2. **Photon→electron**, charge ηE amplified and quantised to 12–14 bits.
3. **White-balance gain:** $R' = g_r R$ etc.
4. **Colour filter array** (Bayer, X-Trans, quad-CFA).
5. **Demosaicking** interpolates missing colours, usually 3×3 or gradient-based.
6. **Colour correction matrix** M maps sensor RGB $\rightarrow$ profile connection space.
7. **Gamma** $\gamma \approx 0.45$ for display.
8. **Sharpening** $I_{\text{out}} = I + \lambda(I - G * I)$.
9. **JPEG quantisation** $Q_{Y,Cb,Cr}$ and entropy coding.

0.5 Camera identification based on photo response non-uniformity (PRNU)

Each pixel responds slightly differently:

$I(x, y) = I_0(x, y) [1 + K(x, y)] + N(x, y)$, where $K(x, y) \sim \mathcal{N}(0, \sigma_K^2)$ captures silicon non-uniformity.

Average K over ≈ 100 flats or natural images yields the camera reference pattern; cross-correlation ρ between query noise residual and K decides *same source?* $\rho > \tau$ match.

0.6 Media forensics based on in-camera fingerprints.

0.6.1 Source identification

Answer *which model/which unit* by matching CFA interpolation kernel, JPEG Q -matrix, radial distortion, on-sensor pattern noise.

0.6.2 Usage of digital fingerprints

- **Identification:** extract fingerprint F , compare to library via $\max_i \rho(F, F_i)$.
- **Forgery detection:** local absence of F or presence of *two* different K -patterns implies splicing.
Bayer periodicity $3.8\mu\text{m}$, proprietary demosaic coefficients, scanner MTF curve, or CG render lack of optical aberration—all discriminate classes.

Tampering detection Clone or replace breaks CFA periodicity; Expectation–Maximisation can localise the doctored block.

Digital Camera Models White-balance gains, colour matrix, proprietary sharpening radii and JPEG Q -tables form a *model* signature even when unit-level PRNU is unavailable.

0.7 Application of geometrical fingerprints in digital forensics.

0.7.1 copy-move

Detect auto-similar patches with $\approx 16 \times 16$ DCT or Zernike descriptors; voting scheme isolates cloned region. Cannot tell originals from copies, but statistical rarity rebuts *bona fide* claim.

0.7.2 Highlights

Eye corneal specular s constrains light direction L via sphere geometry; multiple s_i with divergent normals montage.

0.7.3 Shadows

Under orthographic sun L_∞ and planar ground, shadow edge e gives half-plane constraint $n^\top x \geq n^\top p$; inconsistent wedges betray editing.

0.7.4 Chromatic aberration

Radial colour mis-alignment $\Delta r \approx k_l(r - r_0)$; pasted object with *wrong* Δr or missing purple fringing is suspect.

0.8 Out-camera fingerprints in digital forensics.

Compression, geometric transforms, denoise/sharpen, gamma tweak, Insta-filter—all leave statistically detectable scars that can be dated and ordered.

0.9 Principles of image JPEG compression.

JPEG discards *psychovisually* redundant data.

Redundancy types:

- **Spatial**: neighbouring pixels correlate.
- **Spectral**: Y, Cb, Cr channels correlate.
- **Irrelevance**: HVS masks high freq and colour detail.

0.10 JPEG compression: general compression scheme

Transformer $\mathcal{T}$ (e.g. block-DCT) de-correlates data; Quantiser $\mathcal{Q}$ bins DCT coefficients; Coder $\mathcal{C}$ (Huffman/arith) losslessly packs symbols. Removing $\mathcal{Q}$ gives lossless mode, but ratio $< 2\times$.

0.10.1 Details

- 8×8 blocks $\rightarrow$ DCT $\rightarrow$ each $F(u, v)$ divided by $Q(u, v)$.
- Higher $Q(u, v)$ coarser step larger compression.
- QF rescales entire Q matrix: QF = 100 keeps Q as-is, QF = 50 doubles steps.
- Zig-zag $\rightarrow$ run-length + DPCM DC + VLC.

0.11 JPEG compression: sources of redundancies

Same list framed positively: JPEG removes them; examiner later measures how *much* was removed to detect tampering.

0.12 JPEG compression: particularities of luminance and chrominance channels processing

RGB→YCbCr separates luma Y (perceptually important) from chroma C_b, C_r .

Sub-sample chroma 2:1 horizontal & vertical: $64^2 + 2 \cdot 32^2 = 6144$ bytes vs $3 \cdot 64^2 = 12288$.

0.13 JPEG compression artefacts

Visible: blockiness 8×8 grid, ringing near edges, blur, grain.

Statistical: DCT histogram spikes at multiples of q , zero-run lengths deviate from natural image model—grist for the forensic mill.

0.14 Detection of double JPEG compression

Saving at QF_1 then at QF_2 quantises twice — periodic gaps or heaps in DCT histograms.

Fourier transform of histogram shows strong harmonics when $QF_1 < QF_2$.

First-significant-digit (Benford) distribution also skewed.

At least one of first ten DCT channels peaking — flag *double compressed*.

0.15 Image editing forensics: inconsistency in light

Lambertian diffuse reflectance $L^\top n$ must be consistent for all objects under distant point light; disjoint clusters in L -space signal composite.

0.16 Image editing forensics: local filtering traces

Median filter erases thin lines and forces piece-wise constant patches—unnatural for real sensors.

Detect via predictor residues $\hat{x}_i - \text{median}(W_i)$.

Sharpening by unsharp mask adds overshoot δ at edges; look for gradient histogram tails $|\nabla I| \approx \delta$.

0.17 Photo manipulation with inconsistent shadows: basic assumptions

Treat cast and attached shadows as half-plane constraints on 2-D projection of 3-D light; single L must lie in intersection of all wedges.

0.18 Photo manipulation with inconsistent shadows: cast shadow**0.19 Photo manipulation with inconsistent shadows: attached shadow**

Attached shadow border yields normal n via local gradient; same half-plane rule as cast.

0.20 Photo manipulation with inconsistent shadows: mathematical model

Line through p with normal n : $n^\top(x - p) = 0$.

Half-plane constraint: $n^\top x \geq n^\top p$.

Stack for all shadows: $Nx \geq P$.

Add slack $s \geq 0$: $Nx + s \geq P$; minimise $\sum s_i$.

If optimal $s^* = 0$ all shadows consistent; any $s_i > 0$ tampering.

Solve twice: once with N , once with $-N$, to handle light-behind-camera ambiguity.

0.21 Traces of image editing: resampling detection

Resize/rotate forces interpolation $y = Hx$; periodic dependencies yield spectral peaks in probability map $\mathcal{P}(y_i)$.

FFT of $\mathcal{P}$ exposes the lattice period—evidence of geometric forgery.

0.22 Traces of image editing: detection of global contrast enhancement

Global stretch $T(I) = aI + b$ pumps high-frequency energy; DFT magnitude of residual shows isolated spikes, histogram exhibits gaps/valleys.

0.23 Traces of image editing: detection of local contrast enhancement

Sliding window (sizes $200 \downarrow 20$) repeats global detector; blocks violating model outline the spliced region at progressively finer scales.